ST7305 - Application Note for HSD 2.13” 122x250 Mono TFT-LCD Panel

HSD 2.13” Mono TFT-LCD Panel

1. Introduction

HSD 2.13 ultra-low power display model is a Mono active matrix thin film transistor (TFT) liquid crystal display (LCD) that uses amorphous silicon TFT as a switching device. It is a reflective type display operating in the total reflection and display area with 122 horizontal by 250 vertical pixel resolution (122H x 250V).

2. Application FPC Circuit

1-1. Power Supply

VDDI=VDDA=3.0V (ModeSel=VDDA)

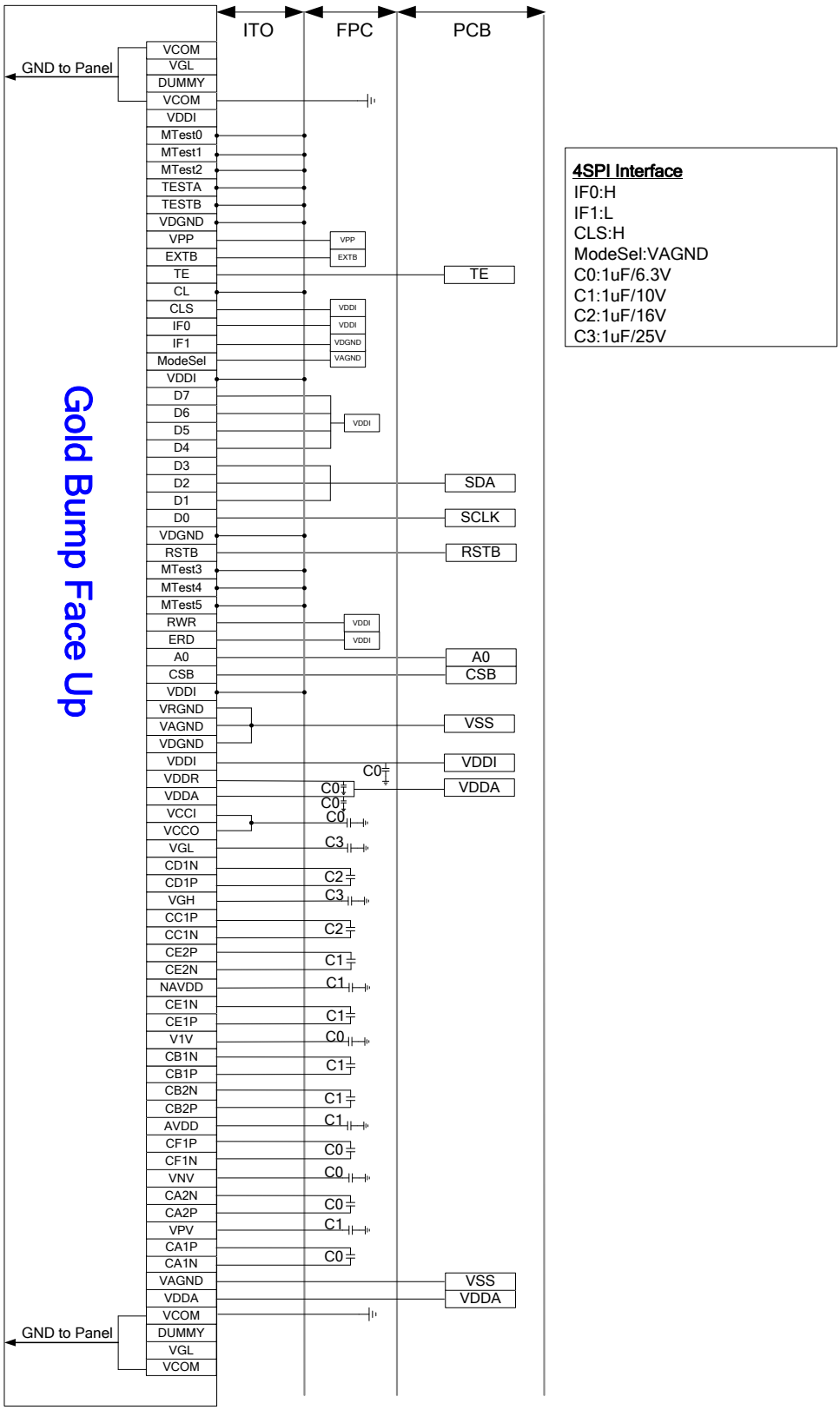
1-2. Interface

IF1	IF0	MPU interface type
L	H	Single 4-line serial interface

1-3. Capacitor

#	Name	Type	#	Name	Type
1	VDDA	1uF/6.3V	11	V1V	1uF/6.3V
2	VDDI	1uF/6.3V	12	CB1N-CB1P	1uF/10V
3	VCCI-VCCO	1uF/6.3V	13	CB2N-CB2P	1uF/10V
4	VGL	1uF/25V	14	AVDD	1uF/10V
5	CD1N-CD1P	1uF/16V	15	CF1N-CF1P	1uF/6.3V
6	VGH	1uF/25V	16	VNV	1uF/6.3V
7	CC1N-CC1P	1uF/16V	17	CA2N-CA2P	1uF/6.3V
8	CE2N-CE2P	1uF/10V	18	VPV	1uF/10V
9	NAVDD	1uF/10V	19	CA2N-CA2P	1uF/6.3V
10	CE1N-CE1P	1uF/10V			

1-4. Application Circuit



3. Reference Initial Code

```
Void ST7305 HSD_Initial_Code(void)
{
    RESET();
    DELAY_ms(100);              //Delay 100ms

    SPI4W_WRITECOM(0xD6);     //NVM Load Control
    SPI4W_WRITEDATA(0X17);
    SPI4W_WRITEDATA(0X02);
    SPI4W_WRITECOM(0xD1);     //Booster Enable
    SPI4W_WRITEDATA(0X01);

    SPI4W_WRITECOM(0xC0);     //Gate Voltage Setting
    SPI4W_WRITEDATA(0X0E);    //VGH=15V
    SPI4W_WRITEDATA(0X05);    //VGL=-7.5V

    SPI4W_WRITECOM(0xC1);     //VSHP Setting
    SPI4W_WRITEDATA(0X41);    //VSHP1=5V
    SPI4W_WRITEDATA(0X41);    //VSHP2=5V
    SPI4W_WRITEDATA(0X41);    //VSHP3=5V
    SPI4W_WRITEDATA(0X41);    //VSHP4=5V

    SPI4W_WRITECOM(0xC2);     //VSLP Setting
    SPI4W_WRITEDATA(0X32);    //VSLP1=1V
    SPI4W_WRITEDATA(0X32);    //VSLP2=1V
    SPI4W_WRITEDATA(0X32);    //VSLP3=1V
    SPI4W_WRITEDATA(0X32);    //VSLP4=1V

    SPI4W_WRITECOM(0xC4);     //VSHN Setting
    SPI4W_WRITEDATA(0X4B);    //VSHN1=-4V
    SPI4W_WRITEDATA(0X4B);    //VSHN2=-4V
    SPI4W_WRITEDATA(0X4B);    //VSHN3=-4V
    SPI4W_WRITEDATA(0X4B);    //VSHN4=-4V
```

```
SPI4W_WRITECOM(0xC5);     //VSLN Setting
SPI4W_WRITEDATA(0X00);    //VSLN1=1V
SPI4W_WRITEDATA(0X00);    //VSLN2=1V
SPI4W_WRITEDATA(0X00);    //VSLN3=1V
SPI4W_WRITEDATA(0X00);    //VSLN4=1V

SPI4W_WRITECOM(0xD8);     //HPM=32Hz
SPI4W_WRITEDATA(0XA6);
SPI4W_WRITEDATA(0XE9);

SPI4W_WRITECOM(0xB2);     //Frame Rate Control
SPI4W_WRITEDATA(0X10);    //HPM=32hz ; LPM=0.25hz

SPI4W_WRITECOM(0xB3);     //Update Period Gate EQ Control in HPM
SPI4W_WRITEDATA(0XE5);    //
SPI4W_WRITEDATA(0XF6);    //
SPI4W_WRITEDATA(0X05);    //HPM EQ Control
SPI4W_WRITEDATA(0X46);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X76);    //
SPI4W_WRITEDATA(0X45);    //
SPI4W_WRITECOM(0xB4);     //Update Period Gate EQ Control in LPM
SPI4W_WRITEDATA(0X05);    //LPM EQ Control
SPI4W_WRITEDATA(0X46);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X77);    //
SPI4W_WRITEDATA(0X76);    //
SPI4W_WRITEDATA(0X45);    //
```

```
SPI4W_WRITECOM(0x62);     //Gate Timing Control
SPI4W_WRITEDATA(0X33);    //0x33=enable Destress; 0x13=disable Destress
SPI4W_WRITEDATA(0X03);
SPI4W_WRITEDATA(0X1F);
SPI4W_WRITEDATA(0X23);

SPI4W_WRITECOM(0xB7);     //Source EQ Enable
SPI4W_WRITEDATA(0X13);    //
SPI4W_WRITECOM(0xB0);     //Gate Line Setting
SPI4W_WRITEDATA(0X3F);    //252 line

SPI4W_WRITECOM(0x11);     //Sleep out
DELAY_ms(120);

SPI4W_WRITECOM(0xC9);     //Source Voltage Select
SPI4W_WRITEDATA(0x00);    //VSHP1; VSLP1 ; VSHN1 ; VSLN1

SPI4W_WRITECOM(0xC7);     //Ultra low power code
SPI4W_WRITEDATA(0xC1);
SPI4W_WRITEDATA(0x41);
SPI4W_WRITEDATA(0x26);

SPI4W_WRITECOM(0x36);     //Memory Data Access Control
SPI4W_WRITEDATA(0X00);    //MX=0 ; DO=0

SPI4W_WRITECOM(0x3A);     //Data Format Select
SPI4W_WRITEDATA(0X11);    //10:4write for 24bit ; 11: 3write for 24bit

SPI4W_WRITECOM(0xB9);     //Gamma Mode Setting
SPI4W_WRITEDATA(0X20);    //20: Mono 00:4GS

SPI4W_WRITECOM(0xB8);     //Panel Setting
SPI4W_WRITEDATA(0X25);    //dot inversion; one line interval; dot inversion

//WRITE RAM 122x250
SPI4W_WRITECOM(0x2A);     //Column Address Setting
```

```
    SPI4W_WRITEDATA(0X19);
    SPI4W_WRITEDATA(0X23);
    SPI4W_WRITECOM(0x2B);     //Row Address Setting
    SPI4W_WRITEDATA(0X00);
    SPI4W_WRITEDATA(0X7C);

    SPI4W_WRITECOM(0x35);     //TE
    SPI4W_WRITEDATA(0X00);    //

    SPI4W_WRITECOM(0xD0);     //Auto power down
    SPI4W_WRITEDATA(0XFF);    //

    SPI4W_WRITECOM(0x39);     //0x39 low power    0x38 high power
    SPI4W_WRITECOM(0x29);     //DISPLAY ON
}
```